

Kaixiang Yao

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EDUCATION

Southern University of Science and Technology

Aug. 2023 - Jun. 2027

B.Eng. in Robotic Engineering; GPA: 3.91/4.00; Rank: 1/91

Advisor: Prof. Chenglong Fu

University of California, Berkeley

Jun. 2026 - Aug. 2026

Visiting Undergraduate Student, Berkeley Summer Sessions; GPA: 4.00/4.00

PUBLICATIONS AND MANUSCRIPTS

[1] Quasi-real-time digital twin for robotic assembly: a physics-based multi-fidelity framework (*Journal of Intelligent Manufacturing, 2026*)

Kaixiang Yao, Hao Chi, Dingye Chen, Jiwen Zhang

[2] RopeFormer: Cross-Trial Adaptation from Interaction History for Dynamic Rope Manipulation (*under review; arXiv:2609.23432, 2026*)

Menglin Wu^{*}, **Kaixiang Yao**^{*}, Shangbo Luan, Masayoshi Tomizuka, Yuxin Chen

^{*} Equal contribution.

[3] TOGEARI: Interaction-Space Preconditioning for Condensed Finite-Element Systems with IPC Contact (*arXiv:2608.14162, 2026*)

Yanlin Liu[†], Chao Huang[†], **Kaixiang Yao**[†], Yao Shen

[†] Equal contribution.

[4] Gaze-Driven Perception-Action Control of Multi-Channel FES for Efficient Grasp Restoration (*revised manuscript under review, IEEE T-MRB, 2026*)

Kaixiang Yao, Zeyang Yang, Jinxin Sun, Kong Hoi Cheng, Junyun Fu, Zhaokai Chen, Chengyu Lin, Chenglong Fu

RESEARCH EXPERIENCE

University of California, Berkeley

Jun. 2026 - Present

Visiting Undergraduate Researcher, Mechanical Systems Control Laboratory

Advisor: Prof. Masayoshi Tomizuka

RopeFormer: Cross-Trial Adaptation from Interaction History for Dynamic Rope Manipulation

- Performed initial simulation validation of trained history-conditioned policies across single-arm swing, bimanual twirl, and transient rope whip tasks.
- Built Newton-to-MJLab sim-to-sim replay and cross-simulator validation for dynamic rope trajectories before hardware deployment.
- Implemented the full Unitree H1-2 deployment stack, integrating dual ZED 2i perception, policy inference, safety constraints, and real-time control.
- Designed and ran all real-world experiments, evaluating frozen policies on unseen physical ropes across repeated trials to measure cross-trial adaptation.

Shanghai Jiao Tong University

Oct. 2025 - Jun. 2026

Undergraduate Researcher

Advisor: Prof. Yao Shen

CARTILAGE: Robot Learning for High-Fidelity Rigid-Soft Contact (ongoing)

- Developed a simulation-and-learning framework for soft-gripper manipulation of fragile objects under high-fidelity rigid-soft contact.
- Designed pneumatic actuation semantics, contact/tactile definitions, and RL-compatible tasks to study force-aware, damage-free grasping.
- Co-developed TOGEARI, an interaction-space preconditioner for condensed finite-element systems with IPC contact, from solver bottlenecks encountered during CARTILAGE.

RESEARCH EXPERIENCE (CONTINUED)

Southern University of Science and Technology	Jul. 2024 - May 2026
Undergraduate Researcher	Advisor: Prof. Chenglong Fu
<i>Eye-Tracking and Electrical Stimulation-Assisted Grasping</i>	
- Proposed normalized gaze-variance features and random-forest recognition with dual-threshold temporal filtering. - Integrated YOLOv5 object detection and MediaPipe hand-keypoint estimation for object-aware grasping strategies. - Designed an 8-channel, 54-electrode high-density FES sleeve for selective muscle activation. - Evaluated the integrated system in 67 grasping trials, achieving 60 successful tasks (90.9%) and 98.4% visual recognition.	

Tsinghua University	Jun. 2025 - Aug. 2025
Undergraduate Researcher	Advisor: Prof. Jiwen Zhang
<i>Digital Twin for Robotic Assembly</i>	
- Built a ROS 2 non-blocking pipeline separating real-time safety control from background FEA. - Used kinematic TIE constraints and mesh-scale analysis to improve robustness and computational predictability. - Integrated full-field stress outputs into an FSM-based safety evaluator.	

TECHNICAL SKILLS

Programming: Python, MATLAB
Robotics and Control: ROS 2, Unitree H1-2, real-time control, sim-to-real deployment
Learning and Perception: Reinforcement learning (PPO), Transformer-XL, ZED 2i stereo vision, YOLOv5, MediaPipe
Simulation and Modeling: Newton, MuJoCo/MJLab, Isaac Lab, randomized dynamics, FEA/FEM, rigid-soft contact
Mechanical Design and Fabrication: SolidWorks, 3D printing, rapid prototyping

HONORS AND AWARDS

First Prize (2nd Overall, Provincial Level)	2025
Guangdong Provincial College Students Engineering Practice and Innovation Competition	
Outstanding Student Scholarship (two consecutive years)	2023 - 2025

RESEARCH INTERESTS

My research interests are in robot learning and embodied AI for physical manipulation, especially contact-rich and deformable tasks, with a growing interest in dexterous manipulation. I am particularly interested in integrating dynamics, contact, force, and embodiment constraints with perception and closed-loop control for reliable adaptation in the physical world.